Regularization

① Lasso (L_1) :-

✓ Feature selection

$$L_1 = \frac{1}{2m} \sum_{i=0}^{m} (\hat{y}^{(i)} - y^{(i)})^2 - \lambda |slop|$$

λ = hyperparameter.

(0.1, 0.001, 0.5)

② Ridge L_2 -

Prevent model from the overfitting

$$L_2 = \frac{1}{2m} \sum_{i=1}^{m} (\hat{y}^{(i)} - y^{(i)})^2 - \lambda (slop)^2$$

3) Elasticnet Reg.

$$EN = \frac{1}{2m}\sum_{i=1}^{m}\left(\hat{y}^{(i)} - y^{(i)}\right)^2 - \lambda(\text{slop})^2 + \lambda|\text{slop}|$$

LR ✓

L_1 ✓

L_2 ✓

EN ✓

☆ polynomial Regression –

PRL → $y = \alpha + \alpha x_1 + \alpha_1 x_2^2 +$

LR = $h_\theta x = \theta_0 + \theta_1 x_1$

$= \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$

Poly $= \theta_0 + \theta_1 x_1$ Simple PLR

$$= \theta_0 + \theta_1 x_1 + \theta_2 x_2^2 + \theta_3 + x_3^3 + \theta_4 x_4^4 + \cdots + \theta_n x_n^m$$